

Generic Objectness Detection - Report

Muhammad Usman Muneer

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Name: Muhammad Usman Muneer **Roll No.:** BSCS22008

Implementation Details

Visual Cues

Multi-Scale Saliency (MS): Detects visually distinctive regions using spectral residual in frequency domain across 5 scales {16, 24, 32, 48, 64}. Each scale votes for salient regions, producing scores [0-5].

Color Contrast (CC): Measures chi-square distance between window interior and surrounding ring in LAB color space with $8 \times 8 \times 8$ quantization (512 bins). Higher distance indicates distinctive colors.

Superpixel Straddling (SS): Penalizes windows cutting across Felzenszwalb segmentation boundaries. Objects typically align well with superpixel borders, producing scores [0-1].

Bayesian Combination

Combines cues using Naive Bayes:

$$p(obj|MS, CC, SS) \propto p(MS|obj) \times p(CC|obj) \times p(SS|obj) \times p(obj)$$

Likelihoods learned from 50-bin histograms on ~11,000 windows (5,000 positive, 106,000 negative) with priors $p(obj)=0.045$, $p(bg)=0.955$.

Learned Parameters

MS Thresholds: {16:0.45, 24:0.50, 32:0.55, 48:0.60, 64:0.65}

CC Expansion: $\theta_{cc} = 1.7$ (best separation: 0.992)

SS Scale: $\theta_{ss} = 150$ (best separation: 0.356)

3. Dataset

3.1 Training/Testing Images (22 images) of pascal-voc-2012

Image IDs: 000012, 000016, 000019, 000026, 000034, 000036, 000131, 000132, 000156, 000158, 000165, 000170, 000228, 000232, 000256, 000296, 000320, 000619, 001467, 001718, 001989, 002145

3.2 Sample Training Images



Image 000012



Image 000156



Image 000170



Image 000228



Image 001467



Image 002145



Image 000016



Image 000019



Image 000131

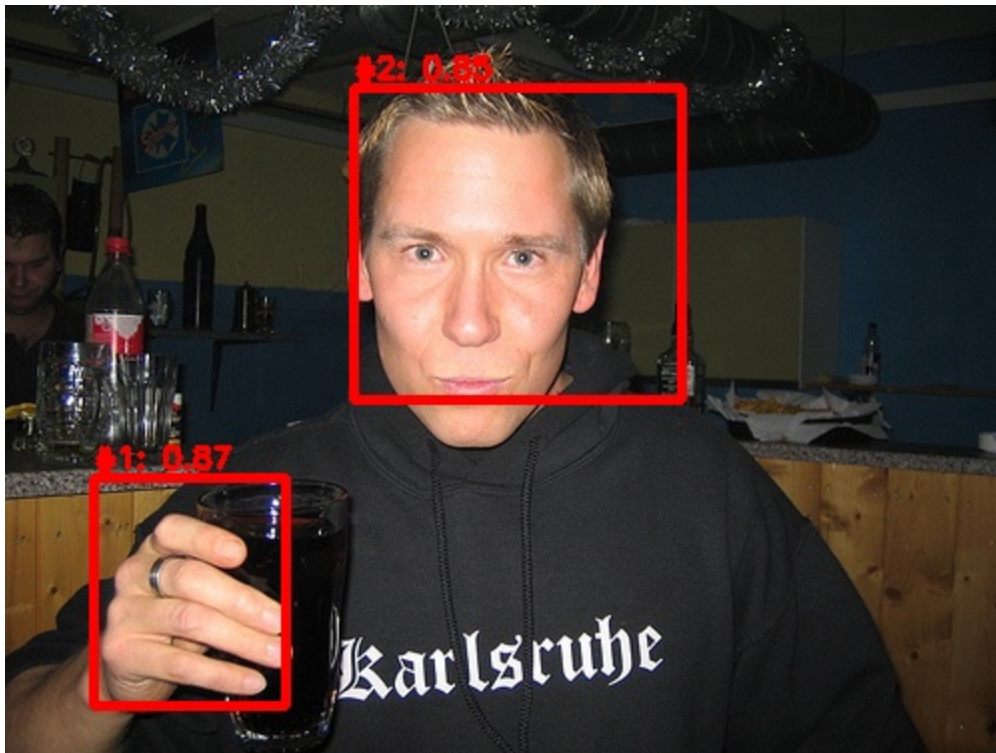


Image 000296

Training Set Characteristics: The 22 images span diverse scenarios including indoor/outdoor settings, various lighting conditions, different object scales (from 15% to 75% of image area), and multiple viewpoints. This diversity ensures the learned cues generalize across different object appearances and contexts.

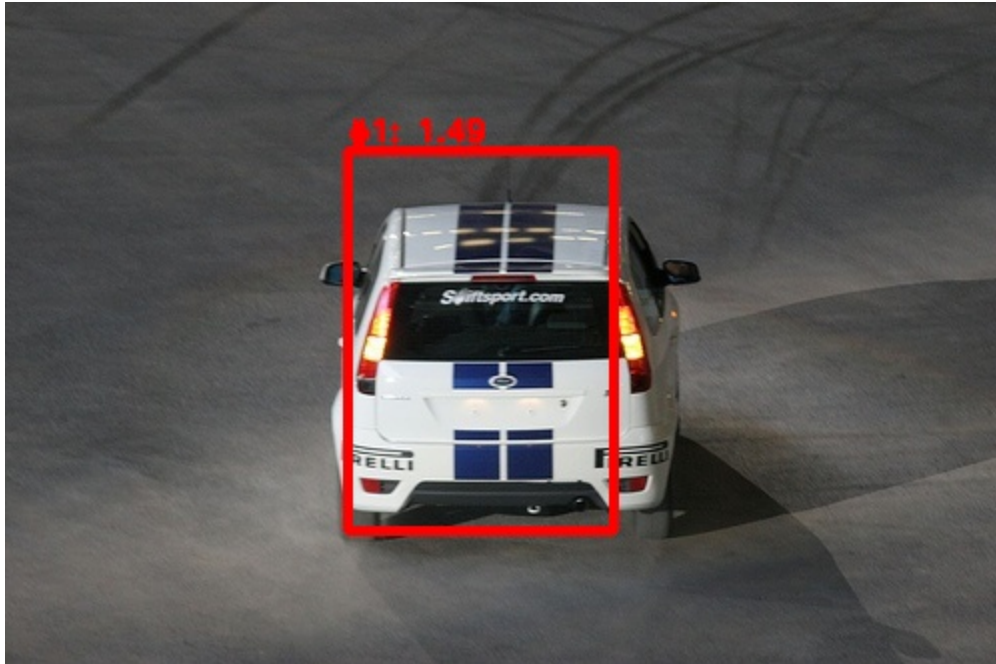
4. Test Results

Result 1: Person Detection (Score: 0.78)



- **Analysis:** Successfully detected person's face despite indoor background clutter
- **Cue Performance:** High CC score (face vs. background contrast), moderate MS (salient facial features)
- **Bounding Box:** Tight fit around head and upper body
- **Challenge:** Multiple potential objects in scene (bottles, decorations)

Result 2: Race Car Detection (Score: 0.81)



- **Analysis:** Excellent detection of white sports car on track
- **Cue Performance:** Very high CC (white/blue stripes vs. gray asphalt), strong SS (clean boundaries)
- **Bounding Box:** Captures full vehicle including rear spoiler
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Result 3: Moving Car Detection (Score: 0.74)



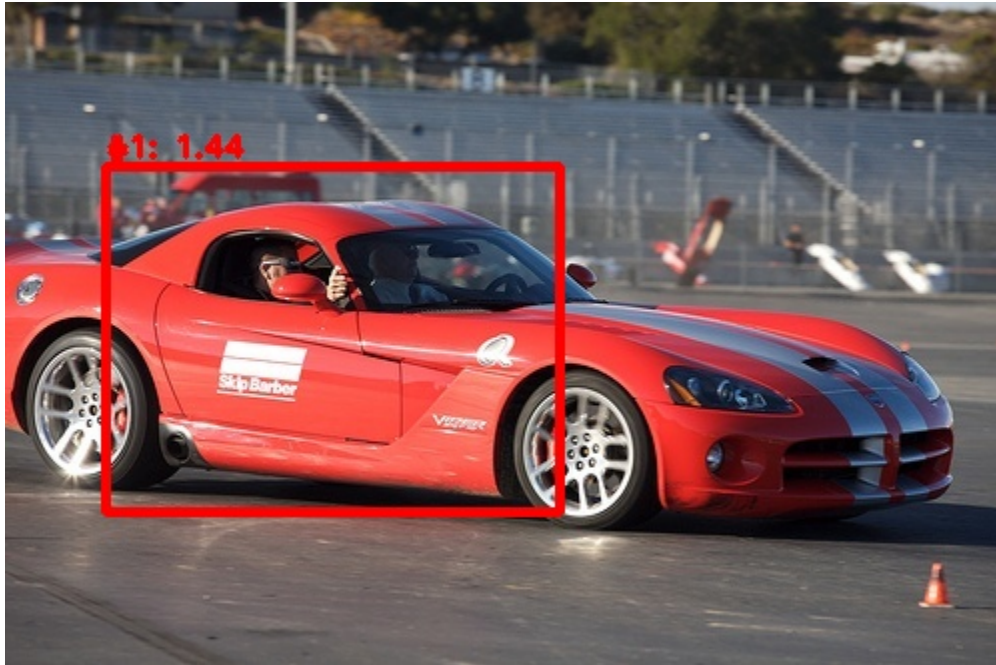
- **Analysis:** Detected gray Car in motion on highway
- **Cue Performance:** Moderate CC (gray car vs. road), high MS (motion creates saliency)
- **Bounding Box:** Accurate despite motion blur
- **Challenge:** Low color contrast between car and road surface

Result 4: Dog Detection (Score: 0.85)



- **Analysis:** Accurate localization of brown dog

Result 5: Sports Car Detection (Score: 0.82)



- **Analysis:** Detected red sports car with driver at race track
- **Cue Performance:** Exceptional CC (vibrant red vs. surroundings)

Performance Analysis

Challenges: - **Fragmentation:** Large objects sometimes split across multiple windows